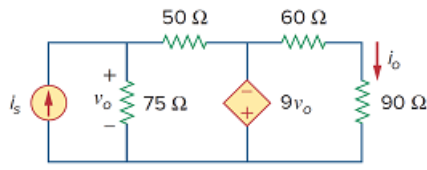


Problem

Calculate the current gain i_o/i_s in the circuit of Fig. 3.105.

Figure 3.105:



Step-by-step solution

Step 1 of 4

Refer to the circuit in Figure 3.105 in the textbook.

Identify the loops and currents in the circuit diagram as shown in Figure 1.

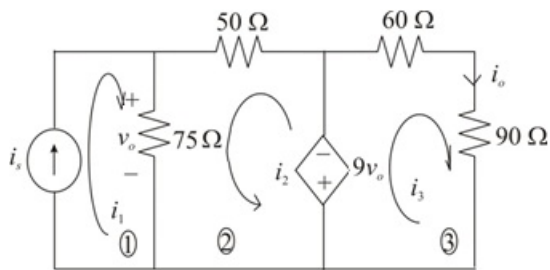


Figure 1

Step 2 of 4

Apply Kirchhoff's Voltage Law in Loop-3.

$$9v_o + i_3(60 + 90) = 0$$

$$9v_o + i_3(150) = 0$$

$$9v_o = -i_3(150)$$

$$v_o = -i_3\left(\frac{150}{9}\right)$$

$$v_o = -i_3\left(\frac{50}{3}\right)$$

Therefore, the expression for the voltage v_o is $-i_3\left(\frac{50}{3}\right)$.

Step 3 of 4

By ohm's law across $75\ \Omega$ resistor, the expression for the voltage v_o is,

$$v_o = (i_1 + i_2)(75)$$

Substitute $-i_3\left(\frac{50}{3}\right)$ for v_o .

$$-i_3\left(\frac{50}{3}\right) = (i_1 + i_2)(75)$$

$$-i_3\left(\frac{2}{3}\right) = (i_1 + i_2)(3)$$

$$-i_3\left(\frac{2}{9}\right) = i_1 + i_2$$

$$i_2 = -i_1 - i_3\left(\frac{2}{9}\right)$$

Therefore, the expression for the current i_2 is, $i_2 = -i_1 - i_3\left(\frac{2}{9}\right)$.

Step 4 of 4

Apply Kirchhoff's Voltage Law in Loop-2.

$$9v_o + i_2(50) + (i_1 + i_2)(75) = 0$$

$$9v_o + 50i_2 + 75i_1 + 75i_2 = 0$$

$$9v_o + 125i_2 + 75i_1 = 0$$

Substitute $-i_3\left(\frac{50}{3}\right)$ for v_o and $\left[-i_1 - i_3\left(\frac{2}{9}\right)\right]$ for i_2 .

$$9\left(-i_3\left(\frac{50}{3}\right)\right) + 125\left(-i_1 - i_3\left(\frac{2}{9}\right)\right) + 75i_1 = 0$$

$$-150i_3 - 125i_1 - 27.778i_3 + 75i_1 = 0$$

$$-177.778i_3 - 50i_1 = 0$$

Replace i_s for i_1 and i_o for i_3 .

$$-177.778i_o - 50i_s = 0$$

$$-177.778i_o = 50i_s$$

$$\begin{aligned}\frac{i_o}{i_s} &= -\frac{50}{177.778} \\ &= -0.2813\end{aligned}$$

Therefore, the value of the current gain $\frac{i_o}{i_s}$ is -0.2813.